Assignment-3

AI-620: Fundamentals of Data Engineering

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Github Link: https://github.com/MDanishZaheer/25280092-AI620-Assignment3

----------Part-1:----------

Task : 1

Already generated both files and stored under "Data" folder root directory.

Task : 2

For Task 2, I have created a expectations for the synthetic dataset using clean dataset statistics as the ground truth that means all the checks applied are based on clean dataset to check the main dimensions of data quality including schema, data types, completeness, uniqueness, validity, and basic distribution. In total 34 expectations defined. and then used it to validate both the clean and corrupted datasets. I have taken clean as a reference and tries to implement all the neccesary expectations that it satisfies.

Following are the expectations defined on each column by me, for each column its name then the defined expectations on it for example sale_id column exists with critical severity written inside paranthesis others are followed in similar way:

sale_id: column exists (critical), type must be int64 (critical), must not be null (critical), must be unique (critical), must be between 1 and 1,000,000 (critical)
customer_id: column exists (critical), type must be int64 (critical), must not be null (critical), must be between 1 and 200,000 (critical)
product_id: column exists (critical), type must be int64 (critical), must not be null (critical), must be between 1 and 20,000 (critical)
sale_ts: column exists (critical), must not be null (critical), must fall within the expected timestamp range (warning)
quantity: column exists (critical), type must be int64 (critical), must not be null (critical), must be between 1 and 219 (critical), quantile distribution check applied (warning)
amount: column exists (critical), type must be float64 (warning), must not be null (critical), must be between 2.04 and 41,533.86 (warning)
channel: column exists (critical), type must be object (warning), must not be null (critical), values must belong to [mobile_app, online, partner, store] (warning)
currency: column exists (critical), type must be object (warning), must not be null (critical), values must belong to [EUR, GBP, PKR, USD] (critical)
Entire dataset: row count must be between 900,000 and 1,000,000 (warning)

Summary of above expectations defined: First I checked schema completeness by verifying that all required columns exist including sale_id, customer_id, product_id, sale_ts, quantity, amount, channel, and currency after that I have defined datatype expectations sale_id, customer_id, product_id, and quantity being integer fields while amount should be numeric decimal and channel and currency should be categorical text fields and then I also added completeness checks so that key columns should not contain null values after that I defined validity rules such as uniqueness of sale_id, valid numeric ranges for IDs, quantity, and amount, allowed values for channel and currency, valid timestamp range for sale_ts, and an expected row count range for the dataset at last I also included a quantile based check on quantity to make sure the distribution stays within the expected behavior.

Data Quality issues: The corrupted synthetic dataset showed multiple important data quality problems that can be observed from the csv report generated and that result in failure of validation check,
1. First issue was schema inconsistency the expected currency column was missing and instead appeared as currency_raw which caused failures in column existence and related checks.
2. Second issue was duplication in sale_id, which should uniquely identify each transaction.
3. The corrupted dataset also had missing values in important fields such as customer_id, product_id, sale_ts, channel, and currency.
4. Datatype problems is also observed because some ID columns are no longer stored in the expected integer format for example customer_id in float instaed on int.
5. There is also many invalid values outside the allowed rules defined. For example, some customer_id and product_id values are outside the expected ranges, some quantity values were zero, negative, or unrealistically high, and amount contained invalid negative, zero, and unusually large values.
6. The sale_ts field also contained missing or out of range timestamps, while channel included invalid category values such as unknown entries.
7. Another issue observe is that the corrupted dataset exceeded the allowed row count limit, which indicates that its overall structure is also affected.

Comparison: Based on the validation comparison between the clean and corrupted synthetic datasets we can observe that the clean dataset passed all 34 expectations successfully because we have defined the expectation suite using it as a ground truth and

give us a overall success as True which means its pass all the defined schemas, values, types, and ranges were all aligned with the defined rules in suite. On the other side the corrupted dataset passed only 16 out of 34 expectations and failed 18 can be checked from the summary report of corrupted data, giving a success as False. This shows that the corrupted dataset had problems across several quality metrics checks. Overall, the clean dataset is reliable and ready for further use while the corrupted dataset clearly requires cleaning and correction before it can be trusted for downstream tasks.

Task : 3

For Task 3, I have created expectations for the PakWheels dataset using the observed dataset structure, domain understanding, and practical rules for used car data thst means the checks applied are based on the important columns and their expected valid behavior and are used to check the main dimensions of data quality including schema, data types, completeness, uniqueness, validity, categorical consistency, and basic distribution. In total, 53 expectations are defined and then used to validate the PakWheels dataset. I tried to implement all the necessary expectations using the data statistics and taken help from the PA2 defined cleaning checkpoints so that the dataset should satisfy them before it can be considered reliable for downstream tasks.

Following are the expectations defined on each column by me. For each column, its name is written first and then the expectations defined on it. For example, for addref, column exists with warning severity is written inside parentheses:
addref: column exists (warning), type must be int64 (warning), must not be null (warning), must be unique (critical)
city: column exists (critical), type must be object (critical), must not be null (critical)
assembly: column exists (warning), type must be object (warning), must not be null (warning)
body: column exists (critical), type must be object (critical), must not be null (critical), values should belong to the allowed body type list (warning)
make: column exists (warning), type must be object (warning), must not be null (warning)
model: column exists (warning), type must be object (warning), must not be null (warning)
year: column exists (critical), type must be float64 (critical), must not be null (critical), must be between 1990 and 2026 (warning)
engine: column exists (critical), type must be float64 (critical), must not be null (critical), must be at least 600 (warning)
transmission: column exists (critical), type must be object (critical), must not be null (critical), values must belong to [Automatic, Manual] (critical)
fuel: column exists (critical), type must be object (critical), must not be null (critical), values must belong to [Petrol, Diesel, Hybrid] (critical)
color: column exists (warning), type must be object (critical), must not be null (warning)
registered: column exists (critical), type must be object (critical), must not be null (critical)
mileage: column exists (critical), type must be int64 (critical), must not be null (critical), must be between 100 and 500,000 (warning), quantile distribution check applied (warning)
price: column exists (critical), type must be float64 (critical), must not be null (critical), must be between 500,000 and 30,000,000 (warning), quantile distribution check applied (warning)
Entire dataset: row count must be at least 50,000 (critical)

Summary of above expectations defined: First I have checked schema completeness by verifying that all required columns exist including addref, city, assembly, body, make, model, year, engine, transmission, fuel, color, registered, mileage, and price after that I have defined datatype expectations where identifier and numeric columns should remain in their correct numeric form while categorical columns should be stored as object type and then I also added completeness checks so that important fields should not contain null values after that I defined validity rules such as uniqueness of addref, valid numeric ranges for year, engine, mileage, and price, allowed values for transmission and fuel, acceptable body categories, and a minimum expected row count for the dataset at last I also included quantile based checks on mileage and price to make sure the distribution stays within a ralistic range.

Data Quality issues: The PakWheels dataset showed multiple important data quality problems that can be observed from the csv report generated and that result in failure of validation checks,
1. First issue was high missing values in some important columns. assembly had 42,928 missing values which is around 69%, body had 7,091 missing values which is around 11 something percent, year had 3,786 missing values which is around 6%, fuel had 709 missing values which is around 1%, color had 1,192 missing values which is around 1%, price had 472 missing values which is around 0.7%, and engine also had 3 missing values only.
2. Second issue was that some important numeric columns contained anomalous or unrealistic values. For example, engine has 151 values below the defined lower bound of 600.
3. Third issue was observed in mileage where 2,453 values were outside the allowed range with unrealistic examples such as 1, 2, 18, 24, 650035, 896554, and 999999.
4. Another major issue was in the price column, where 2,335 values were outside the expected valid range. For example, values such as 250000, 300000, 424000, and 165000 were below the lower bound, while values such as 50000000, 45500000, and 31500000 were above the upper bound, I am considering them as an anomalous and handling them.
5. The price column is also right skewed and contains extreme anomalies.
6. The categorical value checks for transmission, fuel, and body passed, and the quantile checks for mileage and price also passed, the missing values and invalid numeric ranges still show that the dataset is not fully clean.

Validation result analysis: Based on the validation results of the PakWheels dataset we can observe that the dataset passed 43 out of 53 expectations and failed 10, giving overall success as False.

Observations about the reliability and limitations of the dataset: The PakWheels dataset is useful and fairly reliable in overall structure but it is not fully clean or fully reliable without preprocessing. It contains important real world car related features such as city, body, make, model, year, engine, transmission, fuel, color, registered, mileage, and price, which makes it valuable for analysis and future model building that's what make the dataset reliable and useable after preprocessing removing anomalies and all other unreliable data.
The dataset has important limitations, there are high missing values in some columns especially assembly, body, year, fuel, color, and price, which reduces completeness and can affect downstream analysis there are also anomalous and unrealistic values in important numerical columns such as engine, mileage, and price in addition to that the price column is right skewed and contains extreme values, which can affect downstream tasks. So overall the dataset should not be treated as fully reliable in raw form it is better to say that the dataset is practically useful and structurally strong but it still requires cleaning, missing value handling, and outlier treatment before it can be confidently used for conclusions.

Task : 4

For Task 4, I have taken the main data quality issues identified in Task-2 and Task-3 and tried to address the issues that were directly affecting validation and make the data downstream useable. I applied preprocessing on both the corrupted synthetic dataset and the PakWheels dataset so that they will pass the checks defined in expectations, and then validated the cleaned outputs again using the same expectation suites, the task is to remove schema problems fix missing values handle invalid formats and categories remove duplicates where required and filter unrealistic values so that both datasets become more reliable for further downstream tasks.

The data quality issues I chose to address: For the corrupted synthetic dataset, I mainly addressed schema inconsistency, duplicate transaction IDs, missing values in key columns, wrong data types, invalid category values, and values outside the allowed defined rules and the corrupted dataset has currency_raw instead of currency, duplicate sale_id, null values in customer_id, product_id, sale_ts, quantity, amount, channel, and currency, and also invalid values in IDs, quantity, amount, timestamps, and category columns I have resolved all the issues so that the expectations check can be passed.
For the PakWheels dataset, I focused on the important issues observed in Task-3 such as high missing values in assembly, body, fuel, color, year, and price, and anomalous or unrealistic values in engine, mileage, and price. These are the main issues that made the dataset less reliable in raw form.

The cleaning techniques I applied: For the corrupted synthetic dataset, I first converted customer_id and product_id into numeric form, dropped the unnecessary notes column which is blank and renamed currency_raw to currency so that the schema matches the expected structure after that I removed duplicate rows based on sale_id, standardized the text values of channel and currency, converted sale_ts into datetime, dropped rows with missing values in important transactional columns, filled invalid channel and currency values using the mode after standardization, and finally filtered the data using the defined valid ranges for IDs, quantity, amount, and timestamp window.
For the PakWheels dataset, I filled missing assembly values with Unknown, then filtered rows based on practical range rules for price, mileage, engine, and year after that I handled missing body values using grouped mode imputation based on make and model, dropped the remaining rows where body and fuel were still missing, and filled missing color values with Unknown.

The improvements observed in the validation results after cleaning: The improvement after cleaning is clearly visible in both datasets. For the corrupted synthetic dataset, before cleaning it passed only 16 out of 34 expectations and failed 18, with a overall success validation result as False after cleaning the synthetic dataset passed all 34 out of 34 expectations successfully giving overall validation result as True with success. This means that the schema issues, duplicate issue, null value issues, invalid category values, and invalid range issues are successfully handled.
For the PakWheels dataset, before cleaning it passed 43 out of 53 expectations and failed 10, with a overall success of validation result as False. After cleaning, the PakWheels dataset passed all 53 out of 53 expectations successfully, giving overall validation result as True with 100% success. This shows that the missing value problems and invalid numeric range problems were successfully addressed, and the cleaned dataset now fully satisfies the expectation suite.
Overall Task-4 shows that the cleaning techniques applied are effective in improving the reliability of both datasets the cleaned synthetic dataset became fully compliant with the defined schema and expectations rules defined, and the cleaned PakWheels dataset also became fully valid with respect to completeness, datatype, range, uniqueness, categorical checks, and distribution checks so after cleaning, both datasets became much more suitable for downstream related tasks, although this improvement also came with some data reduction because invalid rows had to be removed in order to satisfy the quality constraints.

----------**Part-2:**----------

As per requirement the below is the multiple screenshots attached where in left window the streamlit frontend is running and making prediction while in parallel right window FastApi is running in terminal for both.

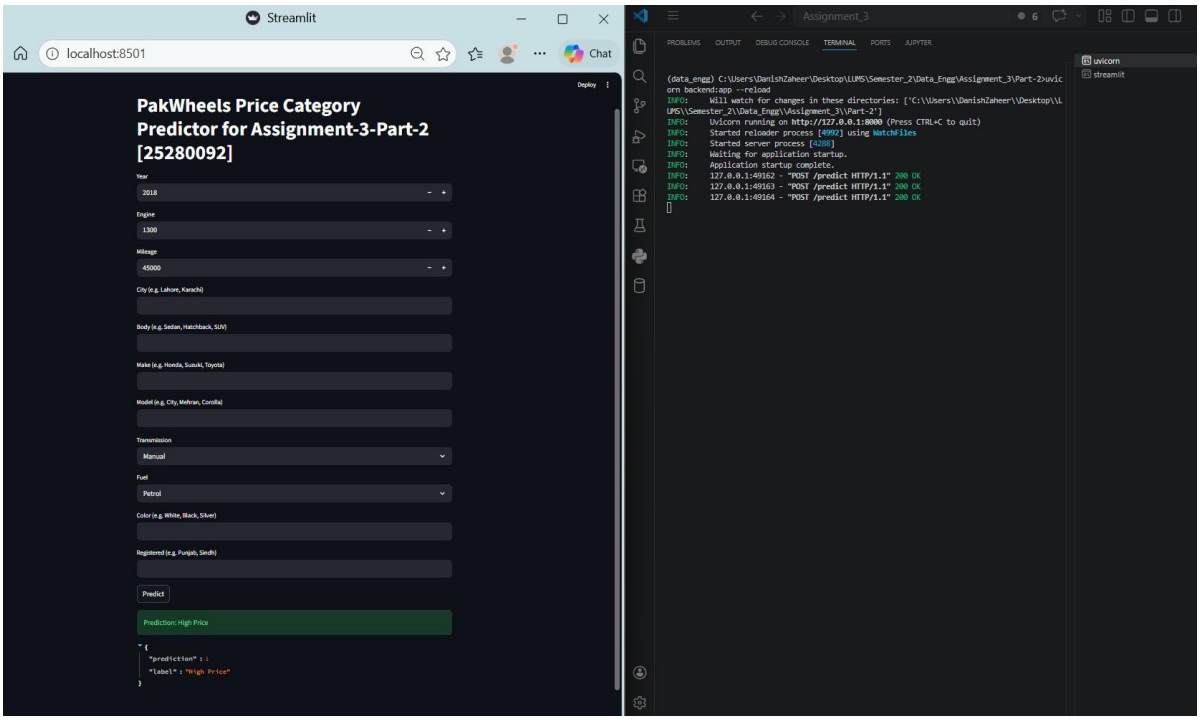

Figure 1: Prediction done on given example assignment pdf file, left is a frontend while right side shows terminal running in background.

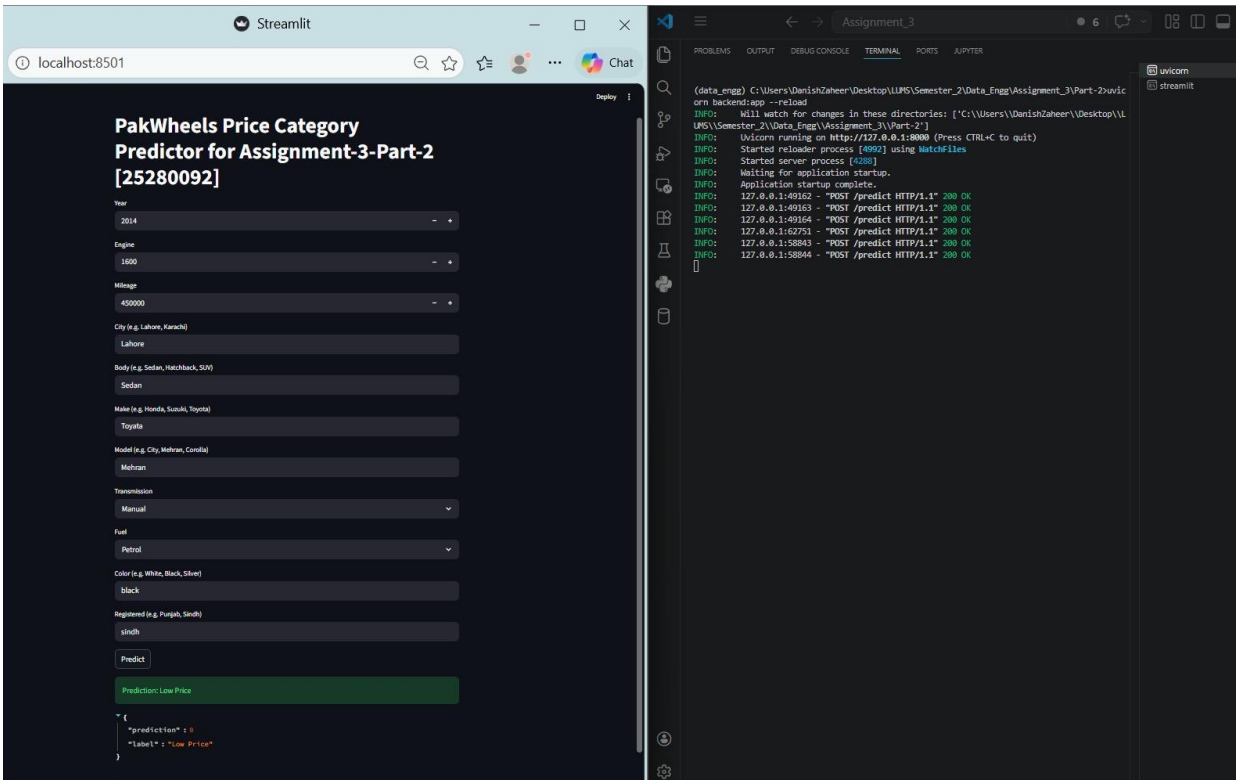

Figure 2: Prediction done on given example assignment pdf file, left is a frontend while right side shows terminal running in background.

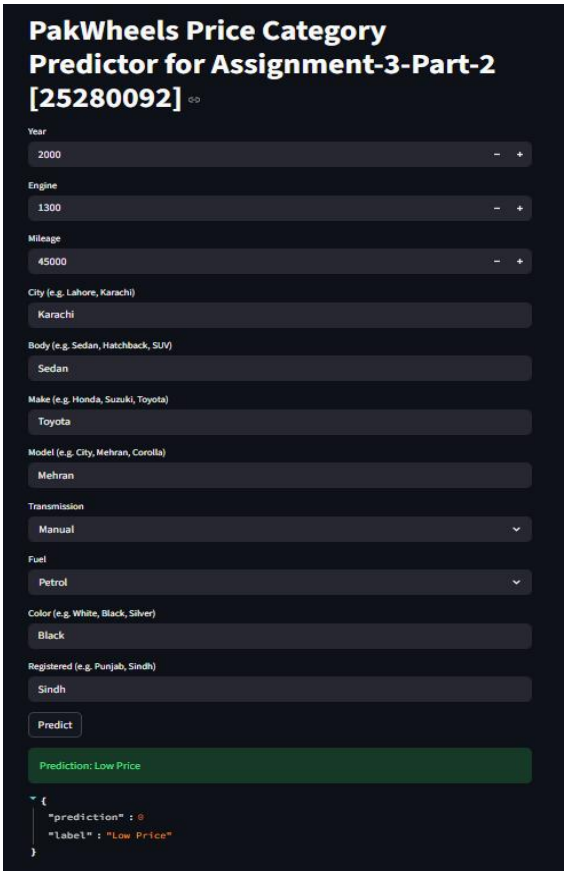

Figure 3: Image showing only Streamlit FrontEnd.

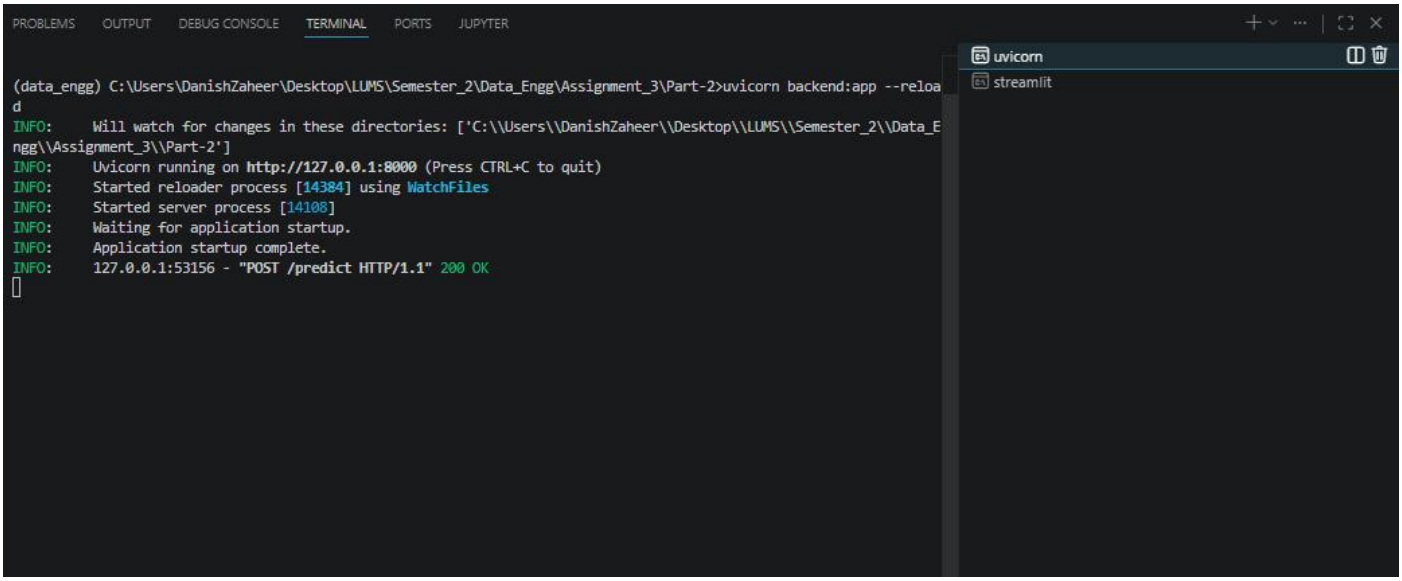

Figure 4: Image showing only FastApi terminal output at backend.